

TECHNICAL UNIVERSITY OF CRETE

School of Electrical & Computer Engineering

Investigating the Applicability of Federated Dynamic Averaging in Federated Low-Rank Adaptation Architectures

Diploma Thesis Presentation

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Background Knowledge : Distributed Learning, FedAvg & Client Drift

- **Federated Learning (FL)**: Privacy-preserving collaborative optimization of foundation models without centralizing client data.
- **Federated Averaging (FedAvg)**:

$$w_{t+1} = w_t + \frac{1}{|S_t|} \sum_{k \in S_t} \Delta_t^{(k)}$$

Clients execute τ local SGD steps on private data $\mathcal{D}_k$.

- **Statistical Heterogeneity (Non-IID)**: Client distributions differ in label, quantity, or features. Modeled via **Dirichlet distribution** $p_k \sim \mathrm{Dir}(\alpha)$.
- **Client Drift**: Divergent local minima pull updates away from the true global optimum.

Dirichlet Heterogeneity (α)

- $\alpha \to 0$: Extreme label skew (clients hold single-class data).
- $\alpha \to \infty$: Uniform IID distribution.
- Key benchmark lever for evaluating drift robustness.

The Dual Communication Bottleneck

- **Spatial**: Model parameter volume (billions of weights).
- **Temporal**: Rigid synchronization cadence ($\tau = E$).

Background Knowledge : Low-Rank Adaptation (LoRA)

- Pre-trained weights $W_0 \in \mathbb{R}^{d \times k}$ are kept **frozen**.
- Updates factorized into low-rank matrices:

$$\Delta W = \frac{\alpha_{\text{lora}}}{r} BA, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \; B \in \mathbb{R}^{d_{\text{out}} \times r}$$

  where rank $r \ll \min(d_{\text{in}}, d_{\text{out}})$.
- Forward pass: $h = W_0 x + \frac{\alpha_{\text{lora}}}{r} BAx$.
- Trainable parameters reduced by $> 99\%$.
- **LoRA Adapter Hyperparameters**:
  - **Rank** $r = 4$: Low-rank bottleneck dimension.
  - **Scaling** $\alpha_{\text{lora}} = 8$: Scaling factor $\frac{\alpha_{\text{lora}}}{r} = 2.0$.
  - **Target Modules**: Query (W_q) and Value (W_v) projections.
  - **Initialization**: Gaussian for A, Zero for B (ensures $\Delta W = 0$ at start).

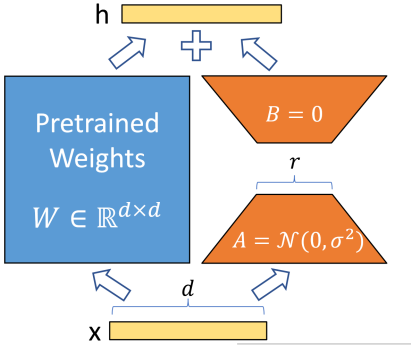


Figure: LoRA decomposition with frozen weights W_0 (Hu et al., 2022).

Background Knowledge : The LoRA Aggregation Inconsistency Problem

Mathematical Non-Commutativity (Bilinear Distortion)

Independent server-side averaging of client low-rank factors fails to reconstruct the composite weight update:

$$\frac{1}{N}\sum_{i=1}^{N}(A_i B_i) \neq \left(\frac{1}{N}\sum_{i=1}^{N} A_i\right)\left(\frac{1}{N}\sum_{i=1}^{N} B_i\right)$$

- Direct FedAvg on (A_i, B_i) induces cross-client geometric interference.
- Destabilizes convergence under non-IID data distributions.
- **Prior Work Limitations**:
  - *Full-matrix expansion* ($\Delta W = AB$): Transmits full parameter matrices $\rightarrow$ destroys spatial savings.
  - *FFA-LoRA*: Permanently freezes down-projection A $\rightarrow$ sacrifices adapter expressivity.
  - *FLoRA*: Server-side SVD recalculation $\rightarrow$ high server compute overhead.

Alternating Coordinate Optimization Restores Aggregation Linearity

RoLoRA resolves the bilinear inconsistency by alternating the active trainable matrix while freezing the other globally across all clients:

- **Odd Rounds (Train B, Freeze A)**: $\frac{1}{N}\sum_{i=1}^{N}(B_iA) = \left(\frac{1}{N}\sum_{i=1}^{N}B_i\right)A = \bar{B}A$
- **Even Rounds (Train A, Freeze B)**: $\frac{1}{N}\sum_{i=1}^{N}(BA_i) = B\left(\frac{1}{N}\sum_{i=1}^{N}A_i\right) = B\bar{A}$
- **Core Limitation**: Enforces a rigid, static local epoch schedule (E), leaving temporal synchronization completely unoptimized.

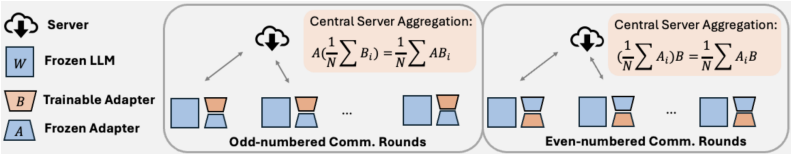


Figure: RoLoRA alternating framework across communication rounds.

Background Knowledge : FDA-Opt & Geometric Monitoring

Theologitis, Samoladas & Deligiannakis, 2025 – Dynamic Variance Halting

- **Geometric Monitoring Root**: Decomposes global threshold functions over distributed streams into local **safe zones** (Sharfman et al., 2006).
- **FDA-Opt Formulation** (Theologitis et al., 2025): Tracks client update dispersion via parameter variance:

$$\mathrm{Var}(W_{t,i}) = \frac{1}{|S_t|} \sum_{k \in S_t} \|\Delta_{t,i}^{(k)}\|_2^2 - \|g_{t,i}\|_2^2$$

  Low variance = aligned progress; high variance = client drift.
- **Event-Driven Halting**: Synchronize when $\mathrm{Var}(W) > \theta_t$.
- **Midpoint Threshold Scaling**: $\theta_{t+1} = \mathrm{ar}_t \cdot \frac{\tau}{2s_t}$.

Research Gap in FDA-Opt

- FDA-Opt operated strictly on **full model parameters** (approximated via AMS sketches).
- Spatial compression (LoRA) and temporal adaptivity (FDA) had never been synthesized.
- Low-rank adapters exhibit high variance volatility requiring decoupled mechanics.

Background Knowledge : Synthesis of Foundational Precursors

Synthesizing RoLoRA (Spatial) and FDA-Opt (Temporal)

Dimension	**RoLoRA** (NeurIPS '25)	**FDA-Opt** (2025)	**FDA-RoLoRA** (**This Thesis**)
Optimization Focus	Spatial parameter volume	Temporal synchronization	**Joint Spatial + Temporal**
Parameter Scope	Low-rank adapters (A, B)	Full model weights	**Low-rank adapters (A, B)**
Aggregation	Alternating coordinates	Standard FedAvg / FedOpt	**Alternating coordinates**
Local Scheduling	Static fixed (E epochs)	Dynamic (Model variance)	**Dynamic (Subspace variance)**
Variance Tracking	None	Full-weight (AMS sketch)	**Exact Subspace** (No sketch)
Threshold Logic	None	Single scalar threshold	**Decoupled Dual-Track** (θ_A, θ_B)

Complementary Synergy

FDA-RoLoRA solves the core limitations of both parent papers: RoLoRA provides the strict aggregation linearity needed for stable variance tracking, while FDA-Opt provides the event-driven cadence needed to eliminate static schedule waste.

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Methodology : Dynamic Averaging in Federated LoRA

Taxonomy of Integration Avenues

Different PEFT architectures have distinct mathematical structures and optimization geometries; dynamic averaging cannot use a single one-size-fits-all rule.

- **FDA + FedAvg-LoRA**: Dynamic averaging with simultaneous (A, B) training.
  - *Drawback*: Inherits bilinear aggregation error; non-linear distortion destabilizes variance monitoring.
- **FDA + FFA-LoRA**: Dynamic averaging with permanently frozen A *[Future Work]*.
- **FDA + SLoRA**: Dynamic thresholding over sparse adapter coordinates *[Future Work]*.
- **FDA-RoLoRA**: **This Thesis**
  - Synthesizes **RoLoRA** (Chen et al., 2025) and **FDA-Opt** (Theologitis et al., 2025).
  - Alternating coordinates provide strict linearity $\rightarrow$ perfect substrate for variance tracking.
  - Maintains independent persistent variance thresholds (θ_A, θ_B) for each subspace.

Methodology : FDA-RoLoRA Architecture

Decoupled Subspace Variance Mechanics

RoLoRA isolates the optimization trajectories of down-projection (A) and up-projection (B). FDA-RoLoRA leverages this to isolate variance tracking and threshold evolution.

- **Active Subspace Mapping**: Server maps active threshold based on designated matrix $\mathcal{M}^{(r)} \in \{A, B\}$:

$$\theta_{\text{current}}^{(r)} = \begin{cases} \theta_A & \text{if } \mathcal{M}^{(r)} = A \\ \theta_B & \text{if } \mathcal{M}^{(r)} = B \end{cases}$$

- **Direct Subspace Variance**: Evaluated exclusively on low-rank coordinate tensors:

$$\text{Var}_{\text{subspace}}(\Delta^{(e)}) = \frac{1}{|S_r|} \sum_{i \in S_r} \|\Delta_i^{(e)}\|_2^2 - \left\| \frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e)} \right\|_2^2$$

- Eliminates memory/compute overhead of mapping drifts back to the full weight space.

Methodology : Independent Halting & Local Training Bounds

Local Epoch Bounds ($e_{\min} = 1$)

Client training bounded in $[e_{\min}, e_{\max}]$ with $e_{\min} = 1$ standardized everywhere to guarantee baseline progress:

$$\text{Halt if } \text{Var}_{\text{subspace}}(\Delta^{(e)}) > \theta^{(r)}_{\text{current}} \ \vee \ e \geq e_{\max}$$

Evaluated at epoch boundaries $e \geq e_{\min}$.

Threshold Controller ($k \cdot e_{\max}$)

Scales threshold toward target horizon $k \cdot e_{\max}$:

$$\theta^{(r)}_{\text{target}} = \left(\frac{k \cdot e_{\max}}{e^*} \right) \text{Var}_{\text{subspace}}(\Delta^{(e^*)})$$

- **Default ($k = 0.5$)**: FDA-Opt midpoint ($e_{\max}/2$).
- **Aggressive ($k = 1.0$)**: Full ceiling target ($e_{\max}$).

Controller Dynamics: Non-IID Success vs. IID Midpoint Lag

Non-IID: Default $k = 0.5$ works almost perfectly containing drift and preventing collapse.
IID: $k = 0.5$ fails to reach the midpoint ($e_{\max}/2$); $k = 1.0$ pushes epochs higher (e.g. toward 20 under $e_{\max} = 40$), but overall struggles against static schedules.

Methodology : Threshold Evolution Mechanics & Controller Design

Primary Operational Controller: Standard Midpoint Scaling

Direct assignment without memory lag: $\theta_{\mathcal{M}^{(r)}}^{(\text{next})} = \theta_{\text{target}}^{(r)}$.

- **Parameter-Free**: No sensitivity to tuning hyperparameters.
- **Adaptive Response**: Responds instantly to changes in gradient drift variance.

Exploratory Temporal Smoothing Variants (Prototyped)

- **Simple Moving Average (SMA)**: Mean over past K active rounds: $\frac{1}{K}\sum_{k=0}^{K-1}\theta_{\text{target}}^{(r-k)}$.
- **Exponential Moving Average (EMA)**: Historical weighting: $\rho\theta_{\text{current}} + (1-\rho)\theta_{\text{target}}$.
- **Finding**: Prototyping revealed smoothing added threshold inertia/lag under distribution shifts without tangible accuracy benefit; hence retained as future work.

Methodology : Coordinate-Wise FedAvg & Protocol Flow

Coordinate-Wise FedAvg Aggregation

- Client drift: $\Delta_i^{(e^*)} = W_{i,e^*}^{(r)} - W_{\text{act}}^{(r)}$
- Cohort average: $\bar{\Delta}^{(r)} = \frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e^*)}$
- Linear FedAvg update ($\eta_s = 1.0$):

$$W_{\text{act}}^{(r+1)} = W_{\text{act}}^{(r)} + \bar{\Delta}^{(r)}$$

- **Linear Aggregation**: Standard FedAvg avoids momentum artifacts, isolating gains strictly to dynamic variance tracking.

Protocol Execution Lifecycle

1. **Toggle**: Server selects active track A or B.
2. **Broadcast**: Sends $W_{\text{act}}^{(r)}$ and $\theta_{\text{current}}^{(r)}$ to S_r.
3. **Train**: Clients run local SGD steps ($\eta_c = 0.1$).
4. **Check**: Evaluate subspace variance at $e \geq e_{\min}$.
5. **Halt**: Return $\Delta_i^{(e^*)}$ when Var $> \theta$ or $e = e_{\max}$.
6. **Aggregate**: Server averages client drifts ($\bar{\Delta}^{(r)}$).
7. **Evolve**: Update active threshold $\theta_{\mathcal{M}^{(r)}}^{(\text{next})}$.
8. **Step**: Server applies FedAvg to active coordinate.

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Implementation : Modular 3-Tier Software Architecture

1. Server Orchestrator

- Global A, B matrices.
- Tracks (θ_A, θ_B) state.
- Coordinates RoLoRA toggle.
- Coordinate-wise FedAvg.

2. Client Nodes

- Private data partitions.
- Local SGD training loops.
- Subspace drift computation.
- Variance checks at $e \geq e_{\min}$.

3. Comm. Interface

- Low-rank serialization.
- PyTorch, HF & PEFT.
- Lightweight FDA-Opt core.

Key Engineering Decisions

- **FDA-Opt Base vs. FederatedScope-LLM**: The RoLoRA paper (Chen et al., NeurIPS 2025) used FederatedScope-LLM without a public release. We adopted and extended the open-source **FDA-Opt** repository (Theologitis et al., 2025), integrating HuggingFace PEFT and RoLoRA coordinate alternation into its modular PyTorch core.
- **Exact Variance over AMS Sketch**: Full-parameter FDA-Opt required randomized AMS sketching to bound memory. In low-rank coordinates ($r = 8$), parameters are so compact that exact subspace variance is computed with negligible overhead, eliminating sketching approximations.

Implementation : Foundation Model Setup & Data Partitioning

Foundation Model Backbone

- **Model**: `roberta-base` (125M params, 12 layers, 768 hidden; selected for multi-seed sweeps under hardware budget).
- **Adapter Placement**: Low-rank adapters injected into Query (W_q) and Value (W_v) projections across top encoder layers.
- **Adapter Configuration**: Rank $r = 4$, scaling factor $\alpha_{\text{lora}} = 8$ (0.47% trainable parameter share).
- **Spatial Reduction**: Backbone frozen; communicating only compact A, B factors cuts parameter transfer by $> 99\%$.

Resource Constraints & Simulation Setup

- **Dirichlet Partitioning**: Client data sampled via $p_k \sim \text{Dir}(\alpha)$ ($\alpha \in \{0.5, 1000.0\}$).
- **25k Dataset Cap**: Datasets capped at $\leq 25{,}000$ samples (MAX_SAMPLES $= 25{,}000$) due to compute limits.
- **Batch Sizes**: $B = 16$ (MNLI, QNLI, QQP), $B = 32$ (SST-2, following Chen et al., 2025).
- **Simulation Early Stopping**: Accuracy-gated early stopping to prevent redundant GPU-hours.
- **Benchmark Tasks**: GLUE suite (SST-2, QQP, QNLI, MNLI).

Implementation : Training Pipeline & Dedicated Testbed Deployment

Training Pipeline Mechanics

- **Alternation**: Track B on even rounds ($t = 0, 2, \dots$), Track A on odd rounds ($t = 1, 3, \dots$).
- **Exact Subspace Variance**: Evaluated centrally at $e \geq e_{\text{min}}$ without high-dimensional projection.
- **Optimization**: Client SGD ($\eta_c = 0.1$), server SGD step ($\eta_s = 1.0$) replicating FedAvg over active coordinate.
- **Orchestration**: Slurm cluster jobs supported; multi-process client worker orchestration.

Polytechnix Server Testbed

- **Server**: SoftNet Lab Polytechnix GPU server.
- **GPUs**: 2× NVIDIA Ampere A10 (24 GB VRAM each).
- **CPUs**: 2× Intel Xeon Silver 4310 (24c/48t, 2.10 GHz).
- **RAM**: 256 GB DDR4 system memory.
- **Comparability**: All 173 runs evaluated exclusively on this uniform platform to ensure perfect comparability.

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Evaluation : Experimental Setup & Benchmark Protocol

Model & Low-Rank Parameterization

- **Backbone**: `roberta-base` (125M params, 12 layers).
- **LoRA Setup**: Rank $r = 4$, $\alpha = 8$ on attention W_q, W_v (0.47% trainable parameters).
- **Uplink Payload**: 240 KB/client/round (sending only active factor A or B).
- **Batch Sizes**: $B = 16$ (MNLI, QNLI, QQP), $B = 32$ (SST-2, following Chen et al., 2025).
- **Optimization**: Mini-batch SGD ($\eta_c = 0.1$, $\eta_s = 1.0$).

Benchmark Matrix & Regimes

- **Tasks**: GLUE suite (MNLI, QNLI, QQP, SST-2) capped at 25,000 training samples.
- **Cohorts**: $C \in \{20, 50\}$ clients (full participation).
- **Regimes**: Homogeneous ($\alpha = 1000.0$) vs. Dirichlet Label Skew ($\alpha = 0.5$).
- **Baselines**: Fixed RoLoRA $E \in \{5, 10, 15, 20\}$.
- **FDA-RoLoRA**: Standardized $e_{\text{min}} = 1$ everywhere; $E_{\text{max}} \in \{20, 40\}$; controller $k \in \{0.5, 1.0\}$.
- **Scale**: **173 completed runs** across 86 configs.

Controller Horizon Dynamics: Default ($k = 0.5$) vs Aggressive ($k = 1.0$)

FDA-Opt default $k = 0.5$ targets midpoint ($E_{\text{max}}/2$). In Non-IID, it worked near-perfectly to halt drift. In IID, $k = 0.5$ lagged behind midpoint (e.g., failed to reach $\bar{e} = 20$ for $E_{\text{max}} = 40$); $k = 1.0$ pushed epochs close to 20, but overall struggled against static baselines.

Evaluation : Executive Performance Scorecard

Comprehensive Synthesis Across GLUE Benchmarks on `roberta-base` ($e_{\min} = 1$)

Task & Cohort	Regime	Top FDA-RoLoRA ($e_{\min} = 1$)	FDA Peak Acc. (%)	Top RoLoRA Baseline	RoLoRA Peak (%)	Δ Acc.
MNLI ($C = 20$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	**76.86 ± 0.07**	RoLoRA ($E = 10$)	76.60 ± 0.60	**+0.26%**
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 1.0$)	**74.47 ± 0.08**	RoLoRA ($E = 10$)	74.24 ± 0.09	**+0.23%**
MNLI ($C = 50$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	**69.85 ± 2.38**	RoLoRA ($E = 10$)	65.85 ± 2.02	**+4.00%**
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 0.5$)	55.56 ± 1.17	RoLoRA ($E = 10$)	**56.55 ± 4.90**	-0.99%
QNLI ($C = 20$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	**83.14 ± 0.24**	RoLoRA ($E = 10$)	81.83 ± 0.65	**+1.31%**
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 1.0$)	**84.22 ± 0.29**	RoLoRA ($E = 10$)	83.50 ± 1.18	**+0.72%**
QNLI ($C = 50$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	82.77 ± 1.51	RoLoRA ($E = 20$)	**88.14 ± 0.00**	-5.37%
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 0.5$)	**75.69 ± 1.17**	RoLoRA ($E = 10$)	74.61 ± 7.18	**+1.08%**
QQP ($C = 20$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	**78.57 ± 2.99**	RoLoRA ($E = 10$)	75.47 ± 0.84	**+3.10%**
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 1.0$)	**77.88 ± 0.46**	RoLoRA ($E = 10$)	77.69 ± 1.54	**+0.19%**
SST-2 ($C = 50$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	89.68 ± 0.80	RoLoRA ($E = 20$)	**92.01 ± 0.46**	-2.33%
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 1.0$)	**85.95 ± 3.96**	RoLoRA ($E = 10$)	59.00 ± 7.86	**+26.95%**

Key Empirical Breakthroughs

Anti-Collapse on SST-2 Non-IID: RoLoRA crashes to 59.00%; FDA secures **85.95%** (+**26.95%** recovery).
Consensus Acceleration on MNLI: Breaks static plateau at 65.85% to **69.85%** (+**4.00%** outperformance).
Communication Reductions: Converges with up to **56.6% lower network bandwidth** (QNLI Non-IID $C = 50$).

Evaluation : Homogeneous Consensus (IID, $\alpha = 1000.0$)

The Compute Accelerator Mechanism in Shared-Gradient Environments

Optimization Dynamics & Controller Behavior

- **Consensus**: Updates align along shared trajectory ($\bar{\nu} \leq 1.0$).
- **Default ($k = 0.5$)**: FDA-Opt $E_{\max}/2$ lagged below midpoint ($E_{\max} = 40$ failed to reach $\bar{e} = 20$).
- **Aggressive ($k = 1.0$)**: Pushed epochs higher ($\bar{e} \to 20$), but struggled overall against static baselines.

Empirical Highlights

- **QQP ($C = 50$)**: Early stops in **14.0 rd** (164 MB) vs 30 rd (352 MB) → **53.3% bandwidth cut**.
- **MNLI ($C = 50$)**: Reaches **69.85**% vs 65.85% plateau (+**4.00**%).
- **QNLI ($C = 20$)**: Peak accuracy **83.14**% vs 81.83%.

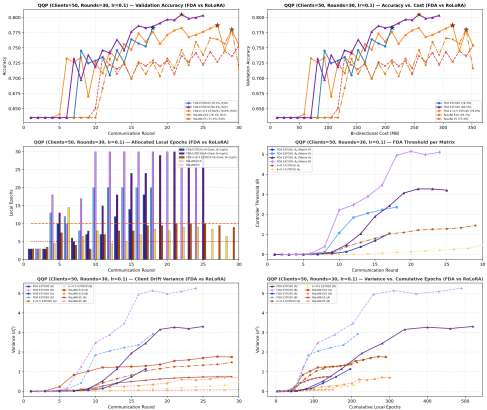


Figure: Multi-panel telemetry on QQP ($C = 50$, IID): (a) Acc/Rounds; (b) Acc/MB; (c) Epochs; (d) θ; (e) $\bar{\nu}$; (f) Var/Epochs.

Evaluation : Statistical Heterogeneity (Non-IID, $\alpha = 0.5$)

The Safety Brake Mechanism & Anti-Collapse Protection Under Label Skew

The "Safety Brake": Near-Flawless Performance

- Under Non-IID, FDA-Opt's controller worked **almost perfectly**.
- Dirichlet skew ($\alpha = 0.5$) causes drift variance to surge **4.0**×.
- Senses spikes and **throttles local epochs**, preventing catastrophic collapse (+26.95% SST-2).

Empirical Highlights

- **SST-2 (**$C = 50$**)**: RoLoRA crashes to 59.00%; FDA reaches **85.95**% (+**26.95**% recovery).
- **QNLI (**$C = 20$**)**: Suite-leading **84.22**% peak accuracy.
- **QNLI (**$C = 50$**)**: Early stops in 13.0 rd (152.3 MB) vs. 30.0 rd → **56.6**% bandwidth savings.

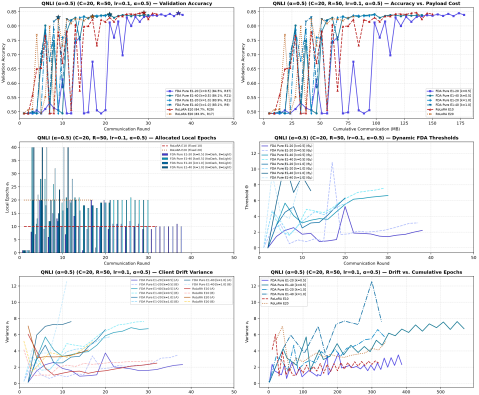


Figure: Multi-panel telemetry on QNLI under Dirichlet skew ($\alpha = 0.5$, $C = 20$): closed-loop tracking securing 84.22% accuracy.

Evaluation : Factor Asymmetry & Decoupled Tracking

Down-Projection (A) vs. Up-Projection (B) Optimization Geometry

Mathematical Factor Asymmetry

- **Down-Projection** $A \in \mathbb{R}^{d \times r}$: Compresses input features into low-rank bottleneck; Gaussian initialization.
- **Up-Projection** $B \in \mathbb{R}^{r \times d}$: Expands bottleneck representations back to hidden dimension; zero initialization.
- Exhibit distinct gradient magnitudes and variance volatility.

Decoupled Threshold Isolation

Maintaining independent thresholds (θ_A, θ_B) ensures that the inactive track remains frozen, eliminating cross-phase geometric interference.

Task & Cohort	Config	Mean $\bar{e}$	$\bar{e}_A$	$\bar{e}_B$
MNLI ($C = 20$, Non-IID)	$E_{\max} = 20, k = 1.0$	17.49	17.19	**17.76**
MNLI ($C = 50$, Non-IID)	$E_{\max} = 20, k = 0.5$	10.13	9.47	**10.80**
QNLI ($C = 20$, Non-IID)	$E_{\max} = 20, k = 1.0$	16.73	16.63	**16.81**
QNLI ($C = 50$, Non-IID)	$E_{\max} = 20, k = 0.5$	10.48	9.83	**11.13**
QQP ($C = 50$, Non-IID)	$E_{\max} = 20, k = 1.0$	14.26	13.83	**14.63**

Key Finding

Up-projection factor B consistently absorbs deeper local epoch allocations ($\bar{e}_B > \bar{e}_A$) across benchmarks, confirming that low-rank factors require decoupled dynamic regulation.

Evaluation : Communication Efficiency: Milestone Gates

Theologitis-Style Analysis: Rounds and Total Payload to Target Accuracies

Task ($C = 20$)	Regime	Methodology	Milestone Gate 1	Milestone Gate 2	Milestone Gate 3	Total MB
SST-2	Non-IID ($\alpha = 0.5$)	RoLoRA ($E = 10$)	R@80%: 23 rd (107.8 MB)	R@85%: 24 rd (112.5 MB)	R@88%: 24 rd (112.5 MB)	117.2 MB
		FDA-RoLoRA ($E_{\max} = 20, k = 1.0$)	**R@80%: 8 rd (37.5 MB)**	**R@85%: 8 rd (37.5 MB)**	**R@88%: 9 rd (42.2 MB)**	**65.6** MB
MNLI	Non-IID ($\alpha = 0.5$)	RoLoRA ($E = 10$)	R@65%: 28 rd (131.2 MB)	R@72%: 43 rd (201.6 MB)	R@74%: 50 rd (234.4 MB)	234.4 MB
		FDA-RoLoRA ($E_{\max} = 20, k = 1.0$)	**R@65%: 19 rd (89.1 MB)**	**R@72%: 25 rd (117.2 MB)**	**R@74%: 36 rd (168.8 MB)**	**178.1** MB
QNLI	Non-IID ($\alpha = 0.5$)	RoLoRA ($E = 10$)	R@80%: 17 rd (79.7 MB)	R@83%: 25 rd (117.2 MB)	R@84%: — (Unattained)	154.7 MB
		FDA-RoLoRA ($E_{\max} = 20, k = 1.0$)	**R@80%: 15 rd (70.3 MB)**	**R@83%: 21 rd (98.4 MB)**	**R@84%: 25 rd (117.2 MB)**	**124.2** MB

2.87× Convergence Speedup (SST-2)

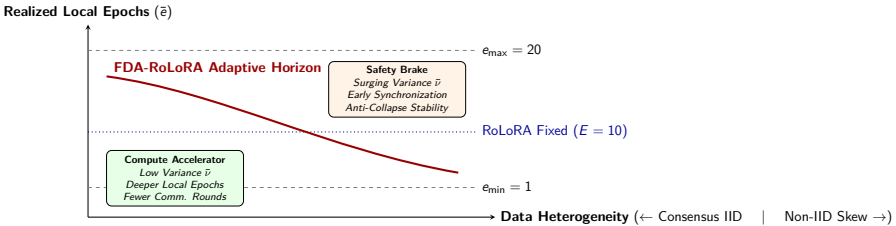
Crosses 80% in **8 rounds** (37.5 MB) vs. 23 rounds (107.8 MB) → **65.2% bandwidth savings**.

Higher Generalization Ceilings (QNLI)

Reaches 84% gate in **25 rounds** (117.2 MB), whereas static RoLoRA fails to ever cross 84%.

Evaluation : Cross-Regime Synthesis: The Dual Frontier

Autonomous Operating Modes Governed by Coordinate Variance



Consensus Accelerator: Low drift → scales $\theta \to \bar{e} \to e_{max}$, slashing round latency.

Anti-Drift Safety Brake: High drift → throttles $\bar{e} \to e_{min}$, preventing catastrophic collapse.

Evaluation : Practical Deployment Guidelines

Four Core Architectural Principles for Real-World Federated Systems

1. Universal Base Progress ($e_{\min} = 1$)

Standardize $e_{\min} = 1$ universally everywhere to ensure baseline progress (exploratory runs testing $e_{\min} = 3$ showed negligible difference).

2. Controller Dynamics in IID vs. Non-IID

- **Non-IID ($k = 0.5$)**: FDA-Opt default ($E_{\max}/2$) works near-perfectly to contain drift and prevent collapse.
- **IID ($k = 1.0$ vs 0.5)**: $k = 0.5$ lags below midpoint; $k = 1.0$ pushes epochs higher (toward 20 for $E_{\max} = 40$), but struggles against static schedules.

3. Decouple Factor Thresholds

Always maintain independent (θ_A, θ_B) states. Freezing the inactive threshold respects low-rank factor asymmetry.

4. Calibrate Early Stopping Under Skew

In non-IID regimes, configure wider smoothing windows ($W \geq 5$) and a round floor ($R_{\min} \geq 0.6 \times R_{\max}$) to tolerate alternating validation oscillations.

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Conclusion : Synthesis of Scientific Contributions

- **Bridging RoLoRA & FDA-Opt**: Unified alternating low-rank adaptation (**RoLoRA**, Chen et al., NeurIPS 2025) with dynamic variance-driven averaging (**FDA-Opt**, Theologitis et al., 2025) into a coherent dual-axis federated framework.
- **Mathematical Consistency via Alternating Isolation**: Alternating coordinate optimization solves the bilinear aggregation inconsistency, providing strictly linear updates that enable stable, distortion-free geometric variance monitoring.
- **Decoupled Subspace Geometric Monitoring**: Evaluating drift variance directly on isolated low-rank coordinate tensors eliminates the memory and communication overhead of full-model gradient approximations.
- **Dual Frontier Operational Model**: Empirically demonstrated across 173 runs on `roberta-base`: acts as a *Compute Accelerator* in consensus (up to 56.6% bandwidth savings) and a *Safety Brake* under skew (+**26.95**% anti-collapse stability recovery on SST-2).
- **Full Empirical Foundation Benchmark**: Completed comprehensive multi-seed benchmark across GLUE tasks on dedicated Polytechnix GPU hardware.

Conclusion : Limitations & Future Research Horizons

Limitations

- **Deterministic Alternation**: Alternates A and B in strict sequence; asymmetric or adaptive switching cadences remain unexplored.
- **Simulated Multi-Process**: Executed on a dedicated multi-GPU server; physical wide-area latency and client dropout left unmodeled.
- **Backbone Scope**: Evaluated on `roberta-base` (125M); multi-billion parameter LLMs (e.g., Llama-3 8B) remain future targets.

Future Horizons

- **Advanced Controllers**: PID-regulated threshold scaling and variance-acceleration damping.
- **Broader PEFT Variants**: Formalizing FDA with FFA-LoRA, SLoRA, and dynamic rank allocation (AdaLoRA).
- **Quantization & Edge**: Synthesizing with 4-bit QLoRA for constrained edge devices.
- **Physical Edge Testbed**: Deployment across heterogeneous physical IoT and mobile hardware.

Key References

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Thank You!

Questions & Discussion

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Open-Source Repository: `https://github.com/Asterinos1/FDA-RoLoRA`